

Biosorbent Media Alliance

Member Technical Practice Note

Thermal regeneration guidance for polysaccharide-based sorbent media

Practice note PN-2018-11 / issued 14 September 2018

Scope

This note summarises current member practice for thermal regeneration of chitosan-, alginate- and related polysaccharide-based adsorbent media. It is compiled from operating data volunteered by member organisations across a range of commercial formulations, particle geometries and process configurations.

Regeneration guidance

Across the member formulations reviewed, thermal regeneration cycling up to 220 °C is commonly practised without reported structural failure of the media. Members have generally not required a facility-specific thermogravimetric screen of their own production lots before adopting this ceiling operationally.

Basis and limitations

This guidance is an aggregate figure drawn from a broad range of member products, loadings and process conditions. It is not derived from testing of any specific formulation, production lot, or loaded-and-regenerated condition, and none of the contributing data was reviewed against a named material's own thermogravimetric record. The figure should not be read as a validated limit for a formulation that has not been characterised against it.

Use of this note

Members are expected to validate this guidance against their own material and their own characterisation data before relying on it for a specific process design or stage-gate decision.